

EVERY DOCUMENT. EVERY MACHINE. ONE AI BRAIN.

FactoryOS AI

Complete Systems Engineering Architecture & Strategic
Specification Blueprint

PRD • Architecture • Database Schema • Knowledge Graph • Agent
Flow • UI Wireframes

Document Type: Comprehensive Product Design & Architecture Document

Engineering Track: Multi-Agent Orchestration & Knowledge Graph Systems

Version: v1.1.0 (Full Technical Artifact Release)

Classification: Internal Enterprise Engineering Specification

1. Product Requirements Document (PRD)

This Product Requirements Document (PRD) outlines the comprehensive functional scope, systemic user personas, core system metrics, and foundational operational guidelines for **FactoryOS AI**—an Industrial Operating System engineered to unify disconnected multi-format legacy datasets into a centralized autonomous Intelligence Platform.

1.1 Core Personas & System Access Matrix

User Persona	Core Problem Solved	Primary System Touchpoint	Access Privileges
Field Technician	Silos of unorganized manuals, slow troubleshooting steps during sudden unexpected failures.	Mobile QR Interface, Ambient Speech Capture Module.	Read asset history, execute shift notes, query troubleshooting SOPs.
Automation / PLC Engineer	Complex debug loops across multi-generational ladder logic codes and functional blocks.	ControlCode Copilot Interface.	Read/Write PLC code configurations, update functional logic maps.
Plant Manager / Safety Officer	Manual verification overhead for statutory compliance tracking (ISO, OSHA updates).	Compliance Guardian Dashboard, Report Studio.	Review safety audits, approve work-order templates, sign off certifications.
C-Suite Executive / CEO	Lack of operational visibility into asset risk profiles, capital expenditures, and financial leakage.	Executive Intelligence Matrix.	Full view of plant performance indices, cost projections, capital allocation.

1.2 High-Level Requirements Matrix

Functional Requirements (FR)

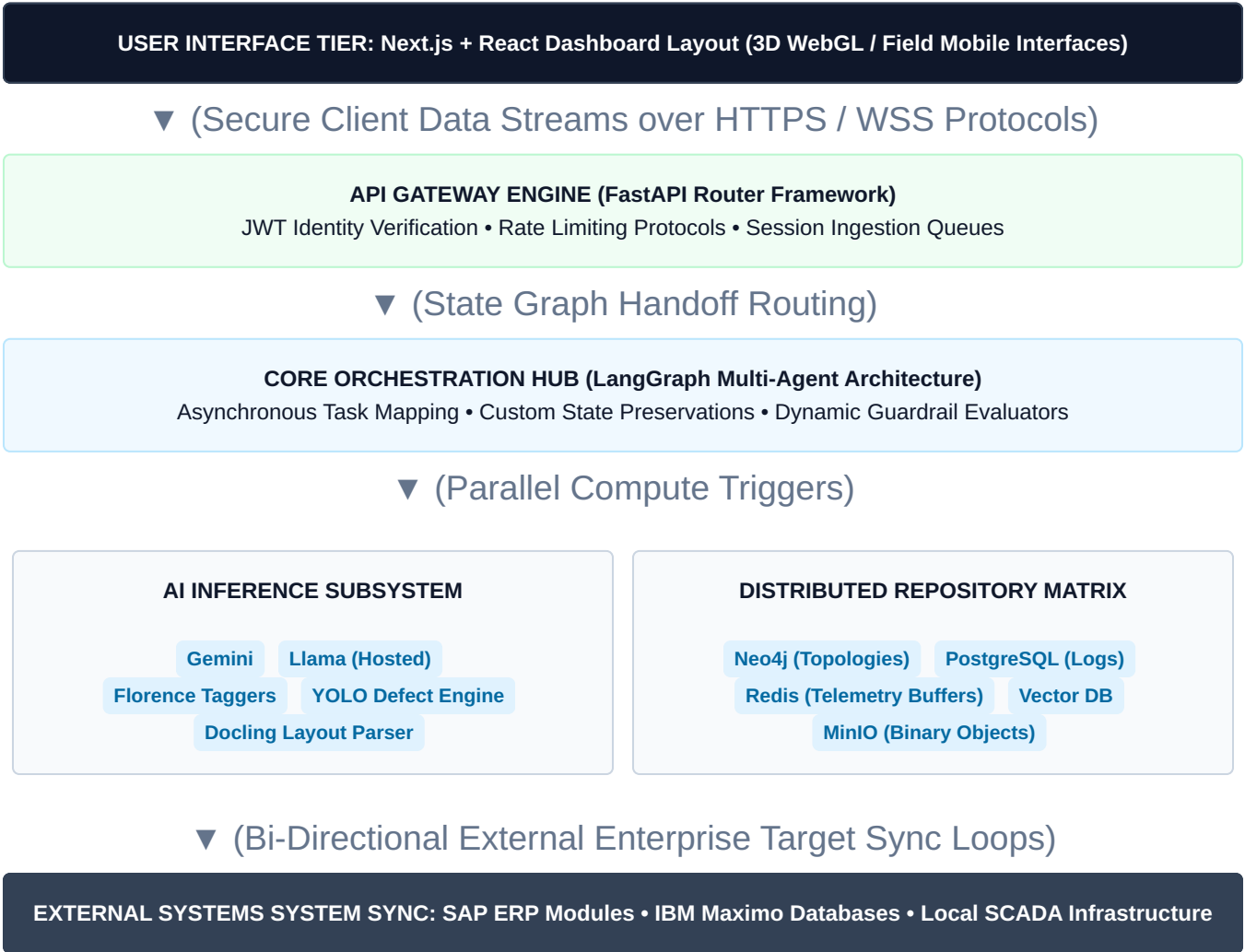
- **FR-101 (Multi-Modal Data Ingestion):** The platform must ingest and process PDFs, structured Excel tables, engineering CAD layers (.dwg), P&ID visual flows, legacy physical manuals, images, and operator emails automatically.
- **FR-102 (Deterministic Vector RAG):** Every conversational extraction layer output must attach direct source verification parameters containing an explicit confidence index, source citation markers, and cross-linked context logs.
- **FR-103 (Autonomous Asset Forensic Tracking):** The system must construct localized event timeline chains automatically tracking historical asset failures back to early parameter drifts.
- **FR-104 (Control Logic Parsing):** The code co-pilot module must parse and analyze Ladder Logic, Structured Text, and Function Block Diagrams to run safety evaluations.

Non-Functional Requirements (NFR)

- **NFR-201 (System Latency Bounds):** Semantic context lookups via the distributed API framework must deliver responses under 1.5 seconds for typical standard natural language queries.
- **NFR-202 (Local Edge Survival):** Local operational nodes must retain local analytical inference capabilities during wide-area network outages, executing high-priority safety analytics completely offline.
- **NFR-203 (Multi-Tenant Graph Isolation):** Relationship data structures inside the Graph Database Layer (Neo4j topologies) must implement strict cryptographic namespace separation across facilities.

2. System Architecture Diagram

The physical system architecture of FactoryOS AI is structured as a decoupled multi-tier operational pipeline. The ingestion flow scales horizontally across independent processing layers, managing raw data normalization prior to agent state orchestration loops.



3. Database Schema

FactoryOS AI handles transactional properties, asset tracking profiles, and system audit history tracking via a highly optimized PostgreSQL relational schema model. Temporal tracking arrays are linked to specialized partition keys to avoid performance degradation over multi-million row historical logs.

3.1 Core Relational Table Definitions

Table: `assets`

Stores the physical equipment taxonomy configuration points representing the layout nodes inside the plant floor.

```
CREATE TABLE assets (  
  asset_id UUID PRIMARY KEY DEFAULT gen_random_uuid(),  
  asset_tag VARCHAR(50) UNIQUE NOT NULL,  
  asset_name VARCHAR(100) NOT NULL,  
  category VARCHAR(50) NOT NULL,  
  location_code VARCHAR(30) NOT NULL,  
  current_status VARCHAR(20) DEFAULT 'OPERATIONAL',  
  remaining_useful_life_days INT,  
  created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,  
  updated_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP  
);
```

Table: `maintenance\_logs`

Captures structural history entries, trace notes, and technical corrective statements executed against specific machines.

```
CREATE TABLE maintenance_logs (  
  log_id UUID PRIMARY KEY DEFAULT gen_random_uuid(),  
  asset_id UUID REFERENCES assets(asset_id) ON DELETE CASCADE,  
  technician_id VARCHAR(50) NOT NULL,  
  incident_type VARCHAR(50) NOT NULL,  
  log_narrative TEXT NOT NULL,  
  confidence_score NUMERIC(4,3),  
  estimated_cost_savings NUMERIC(12,2),  
  logged_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP  
);
```

Table: `compliance\_audits`

Tracks international standards verification parameters alongside dynamic safety audit scores across facilities.

```
CREATE TABLE compliance_audits (  
  audit_id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
```

```
framework_type VARCHAR(30) NOT NULL, -- 'ISO', 'OSHA', 'OISD'
section_reference VARCHAR(50) NOT NULL,
target_asset_id UUID REFERENCES assets(asset_id),
risk_level VARCHAR(20) NOT NULL, -- 'CRITICAL', 'MEDIUM', 'LOW'
audit_score INT CHECK (audit_score BETWEEN 0 AND 100),
status VARCHAR(20) DEFAULT 'PENDING_REVIEW'
);
```

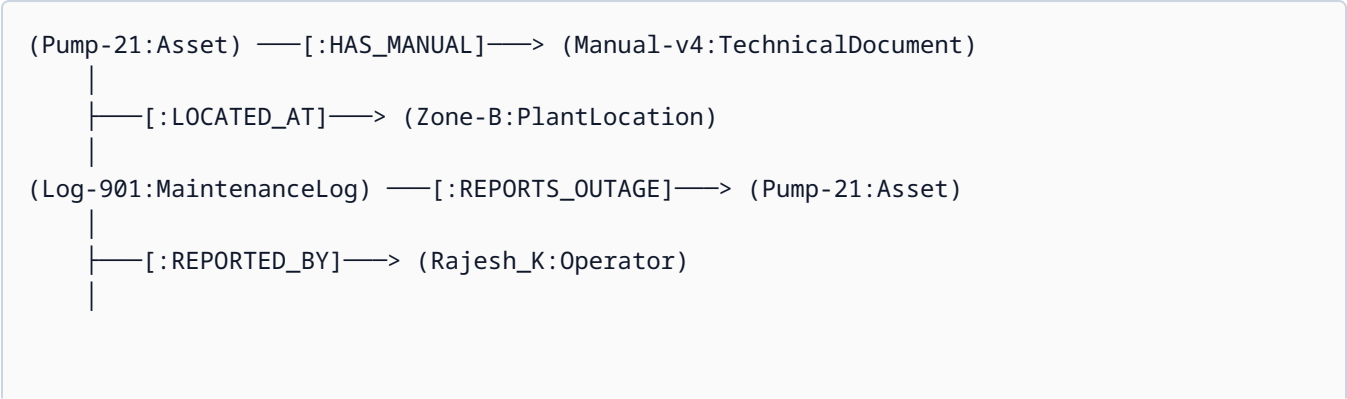
4. Knowledge Graph Design

Unstructured text blocks, operational contexts, physical parameters, and technical procedural cross-references are mapped into a unified semantic **Neo4j Graph Database Model**. This enables deep relational lookups across traditionally siloed datasets.

4.1 Entity Node & Relationship Layout Mapping

Source Node Type (Label)	Relationship Type (Edge)	Target Node Type (Label)	Property Attributes Carried
Asset (`asset_tag`)	`LOCATED_AT`	PlantLocation (`plant_id`)	`distance_meters`, `safety_zone`
Asset (`asset_tag`)	`HAS_MANUAL`	TechnicalDocument (`doc_id`)	`version_hash`, `authoritative_pages`
TechnicalDocument (`doc_id`)	`PRESCRIBES_SOP`	SafetySOP (`sop_code`)	`loto_required`, `revision_date`
MaintenanceLog (`log_id`)	`REPORTED_BY`	Operator (`emp_id`)	`certification_tier`, `shift_code`
MaintenanceLog (`log_id`)	`TRIGGERED_BY`	FailurePattern (`pattern_id`)	`probability_weight`, `root_cause_family`

4.2 Knowledge Topology Visualization Blueprint



└─[:TRIGGERED_BY]─> (BearingFatigue:FailurePattern) ─[:REQUIRES_SOP]─>
(LOTO-09:SafetySOP)

5. AI Agent Flow Diagram (LangGraph Orchestration)

The multi-agent execution network operates on a stateful, cyclic orchestration layout handled via **LangGraph**. Tasks are dynamically assigned to specialized agents depending on the streaming system context modifications.

1. CRITICAL EVENT TRIGGER: Telemetry System Flags Out-of-Bounds Vibration on Boiler-12

▼ (State Context Object Initialized)

2. DOCUMENT AI AGENT (Node Execution)

Pulls context matching manual sections • Queries vector indexing engines for operational boundaries

▼ (State Updated: Document Extracts Loaded)

3. ROOT CAUSE INVESTIGATOR AGENT (Conditional Branch Decision)

Cross-checks historical anomaly trends • Combines sensor data strings via customized temporal scoring frameworks

▼ (Is Probability Validation Score > 85%? If Yes → Route Forward)

4. WORKFLOW AUTOMATION AGENT (Action Node Execution)

Generates Maximo Maintenance Ticket • Auto-reserves Spare Part Profiles • Dispatches Notification Triggers to Certified Techs

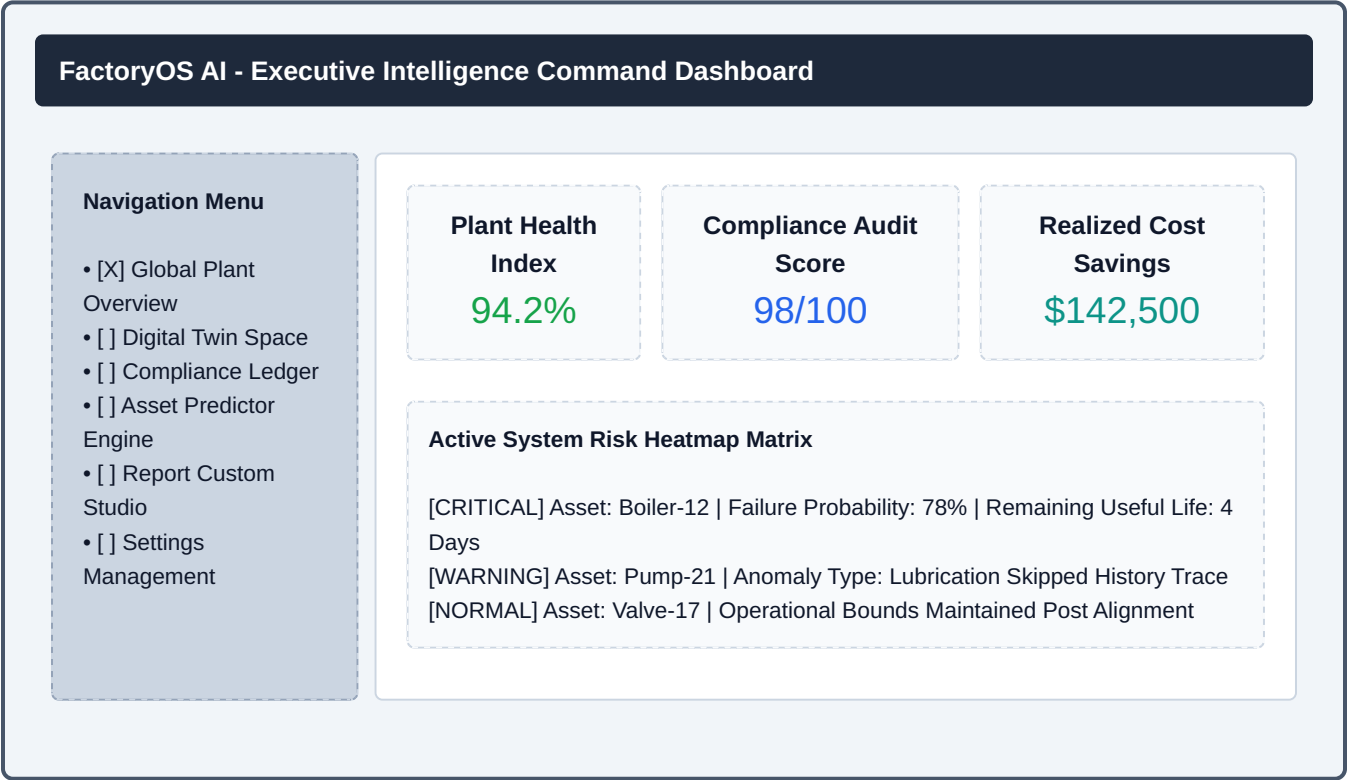
▼ (System Action Completed)

5. SYSTEM LOG CLOSURE: Update Primary Event Store Namespace Matrix & Append to Institutional Memory

6. UI Wireframes Blueprint

This section provides spatial layouts and architectural wireframe designs for key interactive monitoring terminals inside the FactoryOS AI operational ecosystem.

6.1 Enterprise Executive Operations Dashboard



6.2 Industrial GPT & Interactive Digital Twin Viewport Layout

